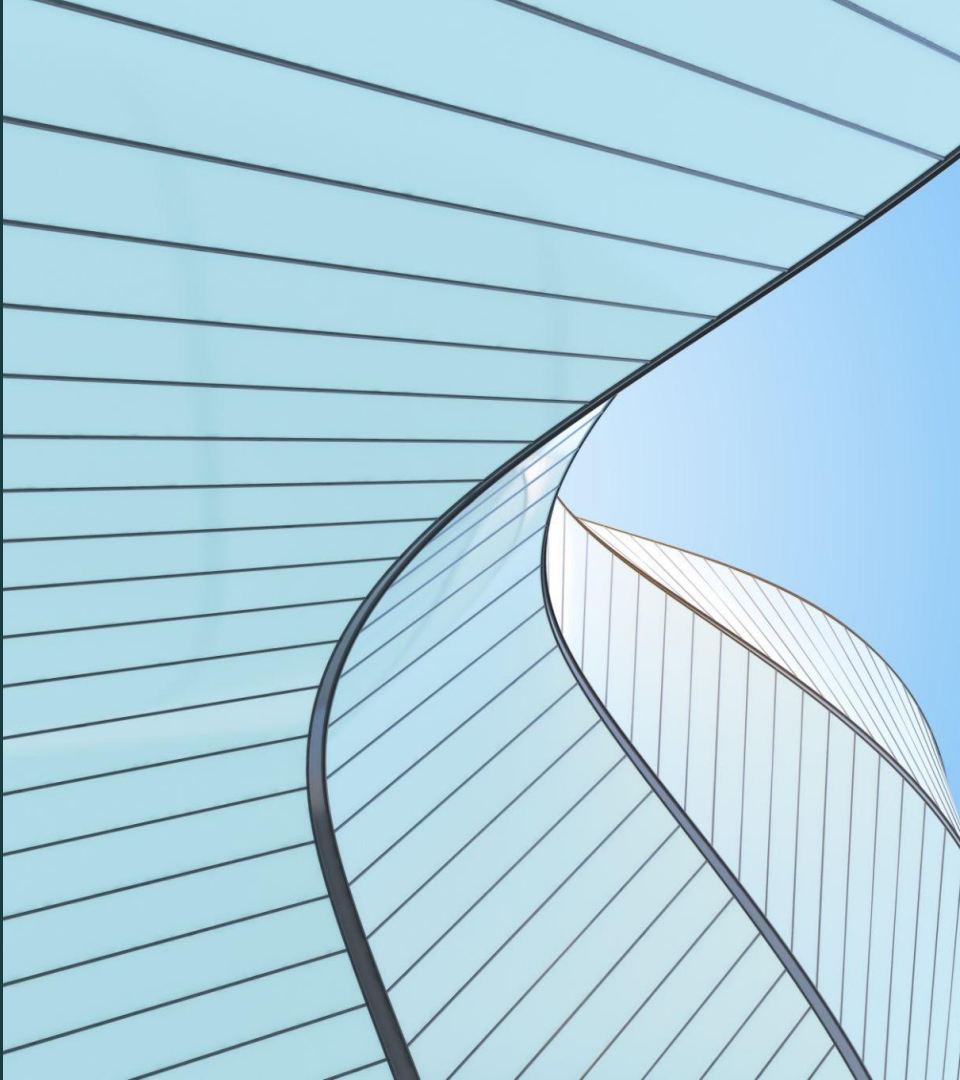


Plagiarism Detection Using Longest Common Subsequence and Jaccard Similarity

By Harry Do and Justin Mei



Content *overview*

Longest Common Subsequence →

Jaccard Similarity →

Implementation →

How to cheat?

Longest Common Subsequence

- Detects shared word order
- Can check for simple rearrangement
- $O(m \times n)$

$$\text{Similarity} = \text{LCS} / \text{MinLen}(A, B)$$

Longest Common Subsequence

A - “The quick brown fox jumps over the lazy dog”

B - “A brown quick fox jumps over the lazy dog”

shared - quick fox jumps over the lazy dog; 66%

Counterexamples

“The quick brown fox jumps over the lazy dog”

“The lazy dog is jumped over by the quick brown fox”

“The agile hazel fox leaps over the resting dog”

Jaccard Similarity

- Measure of similarity between 2 sets
- Time Complexity: $O(|A|*|B|)$
- It is defined as the ratio of the intersection of the sets to their unions:

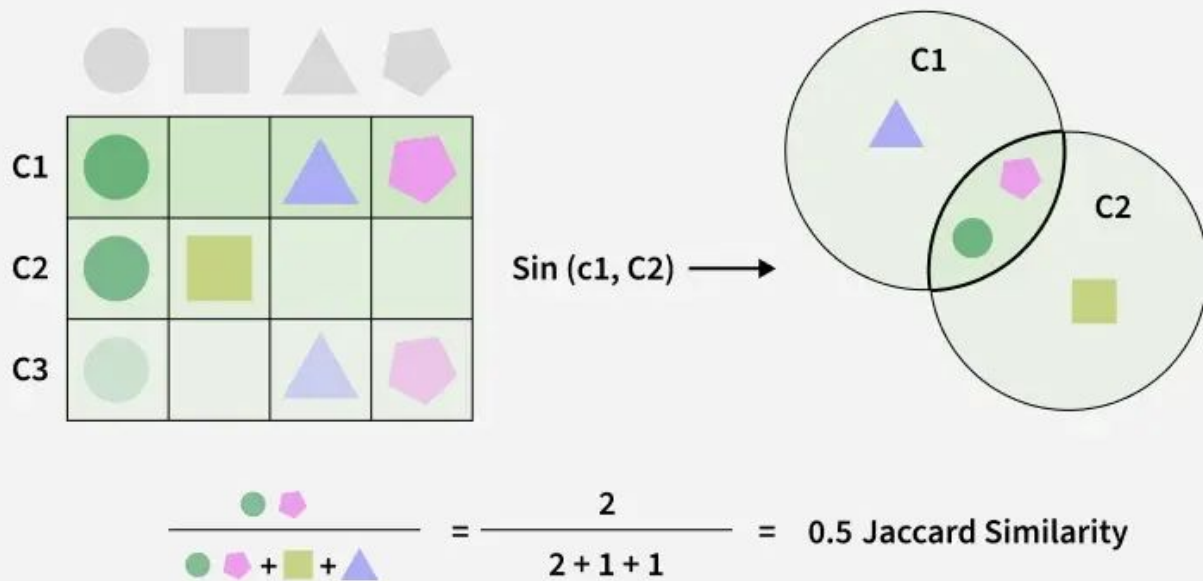
$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

where:

$|A \cap B|$: the number of common elements between sets

$|A \cup B|$: the total number of unique elements in both sets

Jaccard Similarity



Jaccard Similarity

```
text_a = "Generative AI is evolving rapidly."
```

```
text_b = "The field of generative AI evolves swiftly."
```

```
shingles_a = {'enera', 's evo', 'evolv', 'rativ', 'ving ', 'idly.', 'ative',  
'nerat', ' is e', 'is ev', 'olvin', 'i is ', 'pidly', 'ing r', 'rapid', 'apidl',  
've ai', ' rapi', 'tive ', 'gener', ' evol', 'volvi', 'erati', 'ive a', ' ai i',  
'g rap', 'ng ra', 'e ai ', 'lving', 'ai is'}
```

```
shingles_b = {'enera', 'e fie', 'evolv', 'volve', 'wiftl', 'olves', 'rativ', 'f  
gen', 'he fi', ' ai e', ' fiel', 'lves ', 'ield ', ' gene', 'ative', ' swif',  
'nerat', 'es sw', ' of g', 'ftly.', 'ld of', 've ai', 'ves s', 'of ge', 'ai ev',  
'tive ', 'gener', 'the f', ' evol', 'erati', 'iftly', 's swi', 'ive a', 'swift',  
'd of ', 'e ai ', 'i evo', 'field', 'eld o'}
```

Implementation

- 1) Preprocessing the input
 - 2) Tokenization
 - 3) Applying the LCS and Jaccard Similarity
 - 4) Compare the final result
-

Question ?

Thank you



NEXT STEPS

Invite your audience to *learn more* about your organization and to *keep in touch*.

Contact
Full Name
Title
name@example.com

example.com

Our *team*



TITLE
Full Name

Write a brief bio mentioning
their background and area of
expertise.

name@example.com



TITLE
Full Name

Write a brief bio mentioning
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SPOTLIGHT

Full Name

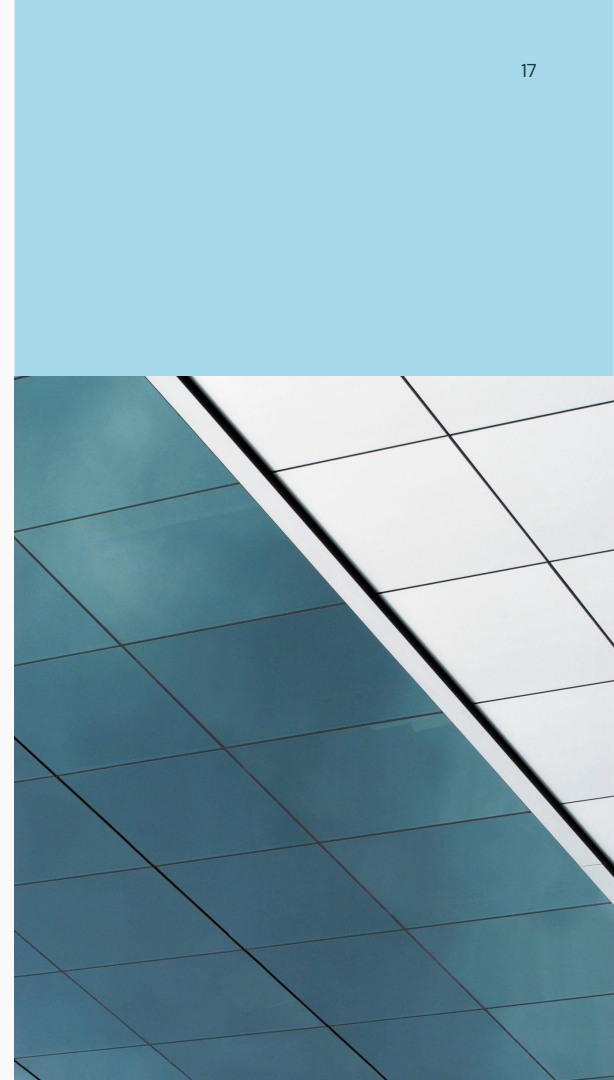
TITLE

Write a *brief bio* mentioning their background and area of expertise.

Contact
name@example.com

This is your mission statement, a chance to *articulate your vision*. Explain what you do, why you do it, and who you do it for. *Be concise. Be confident.*

Add a few words to the statement above. You can explain why this mission is important to you and your company and how you'll achieve it.



OUR MISSION

This is your mission statement, your chance to *articulate your vision*. In a few words, identify what you do, why you do it, and who you do it for. *Be concise. Be confident.*



Add a few words expanding on the above. You can explain why this mission is important to you and your company and how you'll achieve it.

“Add a quote that captures your *mission statement*. Announce the impact you want to achieve and who you aim to help. Show where your *company is heading*, the ultimate goal.”



Full Name
Title

“Add a quote that captures your *mission statement*. Announce the impact you want to achieve and who you aim to help. Show *where your company is heading*, the ultimate goal.”



Full Name
Title



THE PROBLEM

What pain point are you tackling?
What gap are you going to fill?

Briefly describe the impact of the problem and why people should *care*.

THE SOLUTION

Explain what your company or product does, and why it's critical in the current market.

01

VALUE PROPOSITION

Describe a key benefit that your company, product, or service offers to customers.

02

VALUE PROPOSITION

Define what distinguishes your company, product, or service from what's available.

03

VALUE PROPOSITION

Identify your customers and how you'll connect to their behaviors and needs.

04

VALUE PROPOSITION

Copy this slide and add as many items as necessary to fully outline your company's value.

VALUE PROPOSITION

Identify your company's primary benefit to customers and what makes your business or product distinctive.

OBJECTIVES

Write a *confident statement* defining what success looks like to your company.

Introduce a specific business objective

To establish how you'll measure success, identify your key performance indicators (KPIs) and list them like this:

→ KPI 1

→ KPI 2

→ KPI 3

→ KPI 4

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Company Name

AUDIENCE

Write a *keen insight* about your persona or target audience.

Needs and motivations

What does your persona want to achieve? What are their goals?

Pain points

Identify an obstacle your persona struggles with. What interferes with their needs and motivations?

How we can help

Explain where your product or service fits.



Full Name

Introduce your audience persona. Highlight some key traits and mention their age, profession, and location.



Full Name

Age	00
Occupation	Job title
Status	Marital status
Education	Highest degree
Location	Place name
Archetype	Book club

“Quote the persona on their needs and wants. This can be a fictional line or an excerpt from a real user interview.”

Bio

Introduce your persona, including who they are and where they come from. Think about a hypothetical user or customer who represents an audience segment.

Personality



Current emotions

Anxious	Optimistic
Nervous	Enthusiastic

Goals

- What do they want to achieve and why?
- What do they want to achieve and why?
- What do they want to achieve and why?

Frustrations

- What is keeping them from their goals?
- What is keeping them from their goals?
- What is keeping them from their goals?

Favorite brands

LOGO	LOGO	LOGO
------	------	------

COMPETITIVE LANDSCAPE

Write a *high-level insight* about the competitive ecosystem

	COMPETITOR 1	COMPETITOR 2	COMPETITOR 3	COMPETITOR 4
FEATURES	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more	● List the features of this product or service ● 2 or more
STRENGTHS	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more	● Point out the strengths of this product or service ● 2 or more
WEAKNESSES	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more	● Call out the weaknesses of this product or service ● 2 or more

THE OPPORTUNITY

Based on your *competitive analysis* in previous slides, describe the *growth opportunities for your business*. Write a comprehensive claim here, then break it down into individual learnings.

Key learning 1

Go deeper into a specific insight from the competitive analysis and what it means for your business.

Key learning 2

Point to the gap your product will fill and explain how your business can claim that space.

Key learning 3

Consider your brand and customer segments. Explain why you're ideally positioned to reach them.

THE OPPORTUNITY

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Company Name

MARKET ANALYSIS

Provide *key insights* into the state of the market. Share your company's perspective on where the market has been and where it's headed.

Add any additional context about forces or factors influencing the market and how your business is meeting the moment.

00%

Add a few words to explain this metric.

00M

00+

Add a few words to explain this metric.

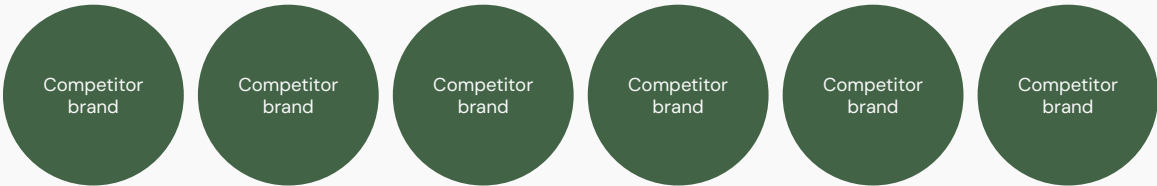
Add a few words to explain this metric.

00M

Add a few words to explain this metric.

Market analysis

Key players



Market value	00
Historic growth rate	00
Projected growth rate	00
Cost of entry	00

Context

Provide some insight on the state of the market today. Describe any challenges, forces, or factors influencing the market. Share your company’s perspective on where the market has been and where it’s headed.

Customer segments



Pricing strategy

PRICE POINT			
MARKET	High	Medium	Low
	Premium		
	Mid-level		
	Economy		

Market analysis

Context

Provide some insight into the state of the market today. Describe any challenges, forces, or factors influencing the market. Share your company's perspective on where the market has been and where it's headed.

Market value	00
Historic growth rate	00
Projected growth rate	00
Cost of entry	00

Key players



Customer segments

Segment 1	Segment 2
Segment 3	Segment 4

Pricing strategy

PRICE POINT			
MARKET	High	Medium	Low
	Premium		
	Mid-level		
	Economy		

BUSINESS MODEL

Clearly outline your business model. This is how you'll create, deliver, and capture value.

Cost structure

What are the major costs of your proposal? Think about salaries, rent, or marketing.

→ Expense 1

→ Expense 2

→ Expense 3

Revenue streams

Outline how you plan to generate revenue through sales, subscriptions, or other methods.

→ Revenue 1

→ Revenue 2

→ Revenue 3

Key activities

What are the core activities your company undertakes to deliver its value proposition? Think about production, research, or fabrication.

Key partnership

Who are your key partners or suppliers? What resources do they provide? And how do they contribute to your business model?

Customer segments

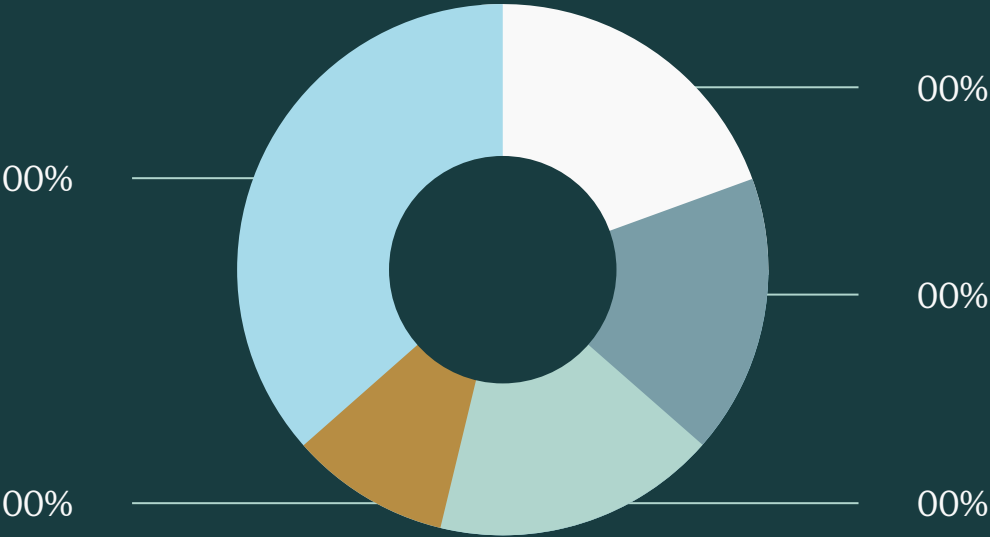
Use this space to restate or prioritize specific customer segments based on your business model.

→ Customer segment

→ Customer segment

→ Customer segment

COST STRUCTURE



- ◆ Staffing
- ◆ Taxes
- ◆ Marketing
- ◆ Logistics
- ◆ Technology

-00%

Estimate potential cost savings in an area or department.

-00%

Add any other potential cost savings in an area or department

If you anticipate changes to your expenses, mention the impact to your cost structure here.

REVENUE MODEL



Revenue stream 1

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 2

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 3

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 4

Define how you'll generate income from this source. Identify relevant pricing or channels.

Revenue stream 5

Define how you'll generate income from this source. Identify relevant pricing or channels.

Company Name

KEY ACTIVITIES

Primary activity *name*

Describe the core activity of your business, how you create and deliver your products or services. This might be manufacturing, product design, software engineering, customer service, marketing, or project management.

Support activity name

Break down your primary activity into the additional activities required to successfully deliver your value proposition. These should contribute to your core business function.

Support activity name

Think about your everyday operations. For example, if your primary activity is software engineering, your support activities may be customer support, software integration, or maintenance.

Support activity name

Show your audience what a day in the life of your organization looks like. Help them understand what's required to do what you do.

CUSTOMER SEGMENTS

Define your target audiences and how you'll speak to them

Target audience 1

Describe a target audience or customer group you want to reach. Identify their traits, values, behaviors, or preferences.

Target audience 2

While your user personas represent ideal individual customers, segments represent a wider range of people who share certain characteristics.

Target audience 3

Think about how each target audience or segment relates to your product or service, why they need it, and how they'll use it.

Target audience 4

Refer back to your user personas and point out which segments they belong to. This will help your presentation feel more interconnected.

Target audience 5

For example, a tech-savvy user persona might fall into a broader Tech Enthusiast or Industry Professional audience segment.

OPERATIONAL PLAN

Announce your vision for scaling operations

STEP TITLE

Define your first goal toward making your business more productive and efficient.

01

STEP TITLE

Identify what you'll focus on next to help your team work together and grow your operations.

02

STEP TITLE

Describe the outcome you want to achieve next. Provide any relevant details about how you'll get there.

03

STEP TITLE

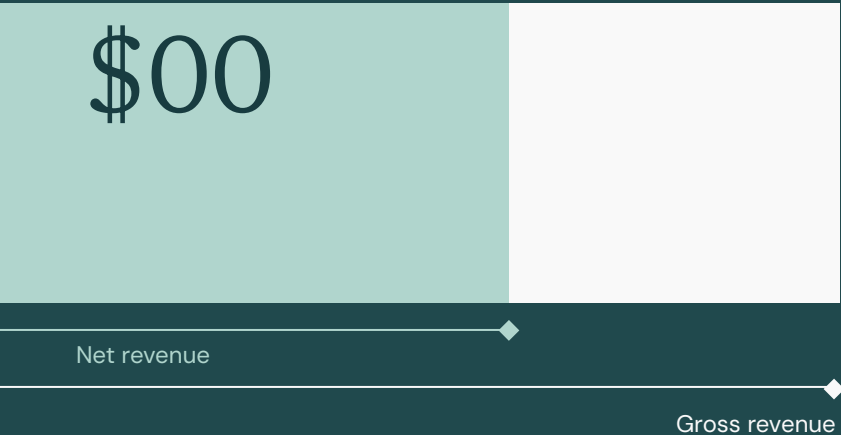
Bring it all together. Share how you'll evaluate, maintain, and advance your operations for long-term sustainability.

04

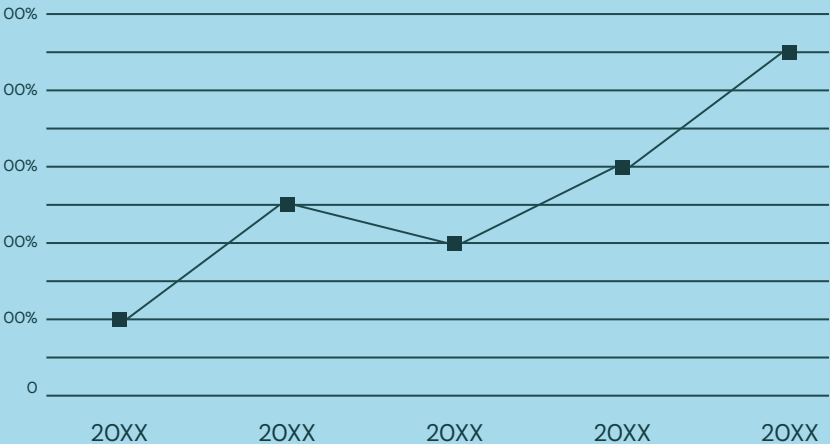
FINANCIAL DATA

Evaluate your revenue trajectory relative to your goals

Yearly revenue



Revenue growth



NEXT STEPS

Invite your audience to *learn more* about your organization and to *keep in touch*.



Contact
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Title
name@example.com

example.com